Unit eight
Heating and Cooling of Electric Motors. Selection of Motor Power Ratings
Thermal model of motor for heating and cooling

We know that when an electrical motor and drive operate, there is a generation of heat inside the motor. The amount of heat generated inside the motor should be known as accurately as possible. That's why thermal modeling of the motor is necessary. The material of the motors and the shapes and sizes of the motors are not unique but the generation of heat does not alter very much depending on these characteristics. So, a simple thermal model of any motor can be obtained assuming it to be a homogeneous body. The main aim of this modeling is to choose the appropriate rating of a motor so that the electric motor does not exceed its safe limit during operation.
At the time 't', let the motor has the following parameters
p_1 -Heat developed, Joules/sec or watts
p_2 -Heat dissipated to the cooling medium, Joule/sec or watt.
m - Mass of the active parts of the machine, kg.
h - Specific heat (the amount of heat required to raise the temperature of 1kg by 1°C; joules/ kg.°C).
A - Cooling Surface, m^2.
L -Co-efficient of heat transfer, Joules/S/m^2/°C.
θ - Mean temperature rise above the ambient temperature, °C.
d θ - Temperature rise in a small interval dt

Now, for time dt, let the temperature rise of the machine be(dθ), Since,
heat absorbed in the machine = (Heat generated inside the machine – Heat dissipated to the surrounding cooling air)
$C(d\theta) = p_1(dt) - p_2(dt)$.....(1) Since, $p_2 = \theta LA = D\theta$.....(2)
Substituting (2) in (1), we get

$$C\frac{d\theta}{dt} = p_1 - D\theta \quad \ldots\ldots\ldots\ldots \quad (3),$$

where,
C=mh - the thermal capacity of the machine in Joules /°C
D=LA- the heat dissipation constant in watts/°C.
Dividing by D yields

$$T_H\frac{d\theta}{dt} = \theta_{ss} - \theta \quad , \text{So, } \theta_{ss} = \theta + T_H\frac{d\theta}{dt} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots (4)$$

where,
$\theta_{ss}=p_1/D$ - steady-state temperature of the machine for t=∞ (final temperature rise above the ambient temperate) .
$T_H=C/D$ - heating time constant.
The first-order differential equation (4) can be solved using the Laplace transformation:

$\frac{\theta_{ss}}{s} = \theta_{(s)} + \left[(sT_H)\theta_{(s)} - T_H\theta_1\right]$, where θ1 is the initial temperature rise above the ambient temperature. Rearranging this equation, we obtain $\frac{\theta_{ss}}{s} = \theta_{(s)}[1 + sT_H] - T_H\theta_1$, then the temperature rise will be

$$\theta_{(s)} = \frac{\theta_{ss}}{s[1+sT_H]} + \frac{T_H\theta_1}{[1+sT_H]} = \frac{a_1}{s} + \frac{a_2}{1+sT_H} + \frac{T_H\theta_1}{1+sT_H} = \frac{\theta_{ss}}{s} - \frac{T_H\theta_{ss}}{1+sT_H} + \frac{T_H\theta_1}{1+sT_H}$$, and finally

$\theta_{(s)} = \frac{\theta_{ss}}{s} - \frac{\theta_{ss}}{\frac{1}{T_H}+s} + \frac{\theta_1}{\frac{1}{T_H}+s}$, using Laplace table, the time function will be

$$\theta_{(t)} = \theta_{ss} - \theta_{ss}e^{-\frac{t}{T_H}} + \theta_1 e^{-\frac{t}{T_H}} = \theta_{ss}(1 - e^{-t/T_H}) + \theta_1 e^{-t/T_H} \quad (5)$$

So, from the above equation we can find out the rise in temperature inside a working machine, which is very near to being accurate and if we plot a graph for the variation of temperature risk with time during heating and cooling and thus the thermal modeling of a motor gets completed. curves 1, 2, 3 and 4 represent

1- $\theta = \theta_{ss}(1 - e^{-t/T_H})$ component of heating curve
2- $\theta = \theta_1 e^{-t/T_H}$ cooling curve from the ambient temperature
3- $\theta = \theta_{ss}e^{-t/T_H}$ cooling curve from the equilibrium point
4- $\theta_{ss}(1 - e^{-t/T_H}) + \theta_1 e^{-t/T_H}$ heating curve

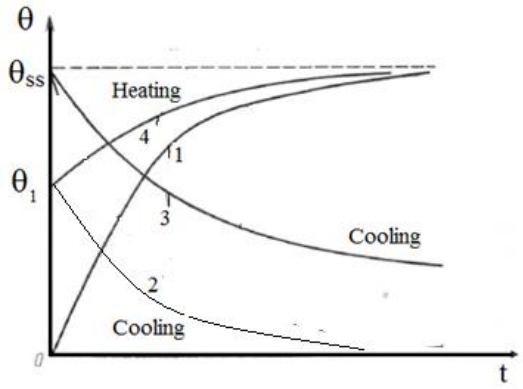


Fig. 1 Heating and cooling Curves

When the temperature rise reaches a constant value, the body is said to have reached maximum temperature rise $\theta_m = \theta_{ss}$, When this condition is reached($d\theta = 0$), The heat developed in the body is completely dissipated to the surroundings which means that no more heat is stored in the motor and the body attains thermal equilibrium.

Let the load on machine be thrown off after its temperature rise reaches a value of θ_2. Heat loss will reduce to a small value (p_1') and the cooling operation of the motor will begin. let the new value of heat dissipation constant be (D'). if the time is measured from the instant the load is thrown off, then

$$C\frac{d\theta}{dt} = p_1' - D'\theta \quad \ldots.(6)$$

Solving this first order differential equation subjected to the initial condition $\theta = \theta_2$ and t=0 gives

$$\theta_{(t)} = \theta_{ss}'(1 - e^{-t/T_C}) + \theta_2 e^{-t/T_C} \quad \ldots (7)$$

where $\theta'_{ss} = \frac{p'_1}{D'}$ is steady state temperature rise for the new condition of operation and $T_c = \frac{C}{D'}$ is a cooling or (thermal) time constant of the machine.

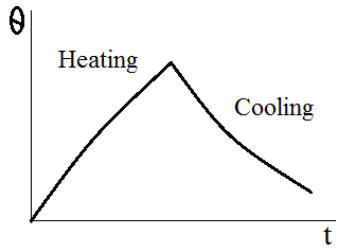


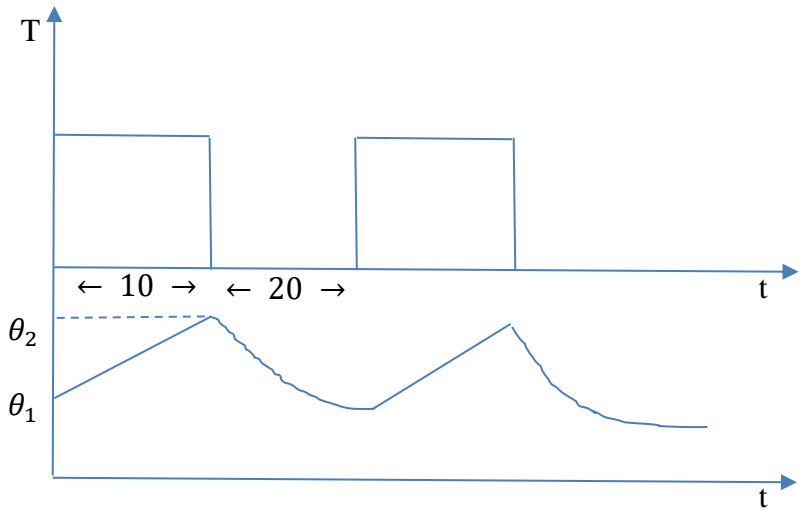
Figure ( ) Heating and cooling curves

→ both heat and cool time constants T_H and T_C depend on the respective heat dissipation constants D & D' which in turn depend on the velocity of cooling air.
→ In self cooled motors, where cooling fan is mounted on motor shaft, the velocity of cooling air varies with motor speed, thus varying cooling time constant T_C.
→ cooling time constant at standstill is much larger than when running.
→ In high performance, medium and high power variable speed drives, motor always provided with separate forced cooling, so that motor cooling be independent of speed.
→ Thermal time constants of a motor are far larger than electrical and mechanical time constants.
→ electrical and mechanical time constants have a typical range of 1 to 100 ms and 10ms to 10s, thermal time constants may vary from 10 min to a couple of hours (بضع ساعات).

Example (4.1):

A motor operates on a periodic duty Cycle in which it is clutched to its load for 10 min and declutched to run on no-load for 20 min. Minimum temperature rise is 40°C. Heating and Cooling time constants are equal and have a value of 60 min. when load is declutched continuously the temperature rise is 15°C . Determine,

1- maximum temperature during the duty cycle, and
2- Temperature when the load is clutched continuously.



Solution:

Given data

motor operated on a periodic duty cycle.
load clutched (or) loaded time (t) = 10 min,
declutched (or) no-loaded time (ta) = 20 min
minimum temperature rise (θ_1) = 40°C.
heating time constant (τ) = 60 min
Cooling time constant (τ') =60 min
when load is declutched continuously,

the temperature rise is (θ'_{ss}) = 15°C.

Maximum temperature during the duty cycle (θ_2) =?

temperature rise when the load is clutched continuously (θ_{SS}) =?

From the diagram we can write

$$\theta_1 = \theta'_{ss}\left(1 - e^{-t_2/\tau`}\right) + \theta_2 . e^{-t_2/\tau`}$$
$$40 = 15\left(1 - e^{-20/60}\right) + \theta_2 . e^{-20/60}$$
$$40 = 15(1 - 0.7165) + \theta_2(0.7165) =$$

4.2525-θ_2 (0.7165). So,

$$\theta_2 = \frac{40 - 4.2525}{0.7165} = 49.9\text{C}^o$$

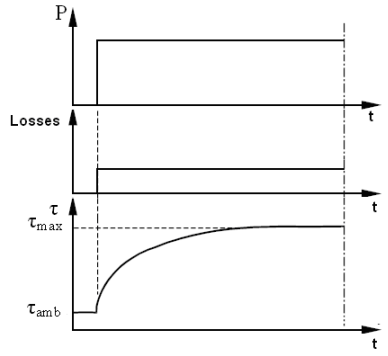
From the diagram we can write

$$\theta_2 = \theta_{ss}\left(1 - e^{-t_1/\tau}\right) + \theta_1 . e^{-t_1/\tau}$$
$$49.9 = \theta_{ss}\left(1 - e^{-10/60}\right) + 40 * e^{-10/60}$$
$$49.9 = \theta_{ss}(1 - 0.8464) + 40(0.8464) = \theta_{ss}(0.1536) + 33.856$$
$$\theta_{ss} = \frac{49.9 - 33.856}{0.1536} = 104.45\text{C}^o$$

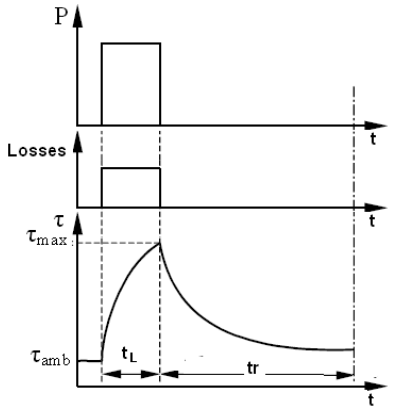
The maximum temperature during the duty cycle $\theta_2 = 49.9\text{C}^o$
Temperature when the load is clutched continuously $\theta_{ss} = 104.45\text{C}^o$

Motor Duty Type (The eight - S1 - S8 - IEC duty cycles of operating electrical motors):

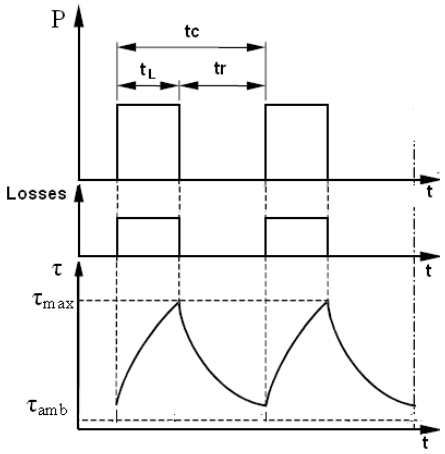
IEC (the International Electrotechnical Commission) uses eight duty cycle designations to describe an electrical motors operating conditions (Standard types of duty classes):
These types describe how the load and motor output varies with time. The motor must undergo a load test without exceeding the temperature limits.
Continuous time duty (duty type S1) : The motor works at a constant load for enough time to reach temperature equilibrium. This duty is characterized by a constant motor loss. Examples: Paper mill drives, conveyers, compressors, centrifugal pumps, fans etc.



Short time duty (duty type S2): The motor works at a constant load during a given time, but not long enough to reach temperature equilibrium. The rest periods are long enough for the motor to reach ambient temperature. The recommended values for the short time duty are 10,30,60 and 90 minutes. e.g. Cranes, valve drives, drives for position control, gates, doors… etc



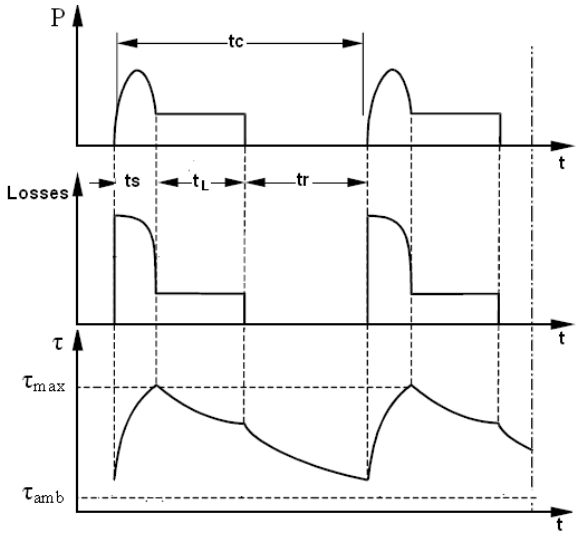
Intermittent periodic duty (duty type S3): Sequential, identical run and rest cycles with constant load. Temperature equilibrium is never reached because these periods being too short to attain thermal equilibrium during one duty cycle. Starting current has little effect on temperature rise. The duration of the duty cycle is 10 minutes. The recommended values for the cyclic duration factor (CDF) are 15, 25, 40 and 60 percent. Where CDF= $\frac{t_L}{t_c}.100\%$. Examples: motors that drive hoisting mechanisms, trams and trolley buses, metal cutting and drilling tool drives, fork lift drives, pressing machine drive… etc.



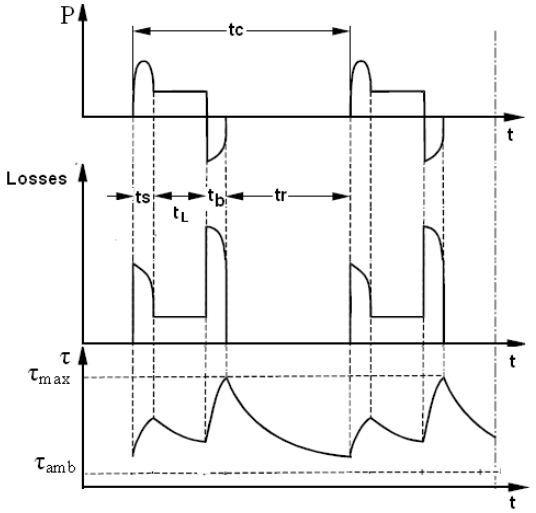
Intermittent periodic duty with starting (duty type S4): Sequential, identical start, run and rest cycles with constant load. These periods being too short to attain thermal equilibrium during one

duty cycle, but starting current affects temperature rise. In this duty, the stopping of the motor is obtained either by natural deceleration after disconnection of the supply or by means of braking such as a mechanical brake which does not cause additional heating of the windings.

CDF= $\frac{t_L + t_s}{t_c}$.100%=15, 25, 40 and 60%. Number of starting (cycles) per hour C/h=30,60,120 and 240. Factor of inertia (FI) = $\frac{J_\Sigma}{J_m}$=1.6, 2.5, 4, 6.3 and 10 where, J_Σ is the referred inertia to the motor shaft, J_m is the motor inertia, inertia is measured by kg×m^2. Examples: motors driving metal-cutting lathes, drives for fork lift trucks (قيادة الشاحنات ذات الرافعات الشوكية), mine hoist (رافعة المناجم) …etc.



Intermittent periodic duty with starting and braking (duty type S5) : A sequence of identical duty cycles, each cycle consisting of a starting period, a period of operation at constant load, a period of rapid electric braking and rest period. The duty cycle is too short for thermal equilibrium to be reached. CDF= $\frac{t_L + t_s + t_b}{t_c}$.100%=15, 25, 40 and 60%. Factor of inertia (FI) =2 , 6.3, 10.

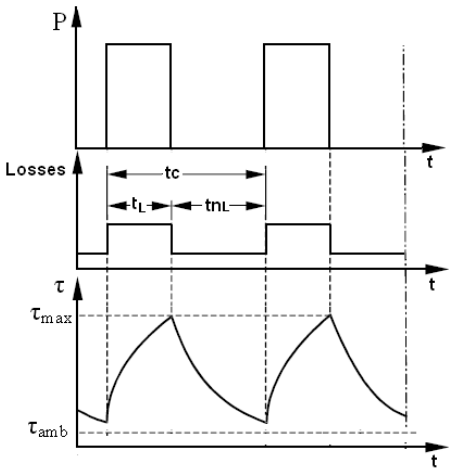


Billet mill drive (قيادةمصنع القضبان), manipulator drive(قيادة مناولة المواد), ingot buggy drive(قيادة قوالب عربات الجر), several machine tool dives, drives for electric suburban trains(قطارات الضواحي) and mine hoist are some of examples of this duty.

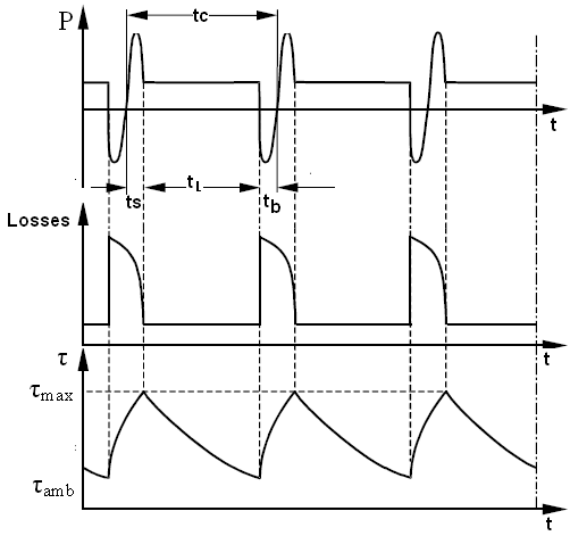
Note: In (S3 to S5) types of duties, the loading period and rest period of the cycle are so short that the steady state temperature is not reached. While working on these identical cycles continuously, the motor temperature rise must not exceed the permissible values for a given class of insulation. Crane duty applications can be classified into duty types S2 to S5.

Continuous operation with intermittent load (duty type S6) : Sequential, identical cycles of running with constant load and running with no load. No rest periods. The periods are too short for thermal equilibrium to be obtained during one duty cycle.

CDF= $\frac{t_L}{t_c}.100\%$ =15, 25, 40 and 60%. Pressing, cutting, shearing and drilling machine drives are the examples.



Continuous operation with periodic starting and braking (duty type S7): Sequential identical cycles of starting, running at constant load and electric braking (e.g. counter-current braking). No rest periods. The periods are too short for thermal equilibrium to be obtained during one duty cycle. This class of duty is indicated by:
C/h=30,60,120 and 240, FI =2 , 6.3, 10. an example is motors driving blooming mills(مدلفنة نورات الحديد).

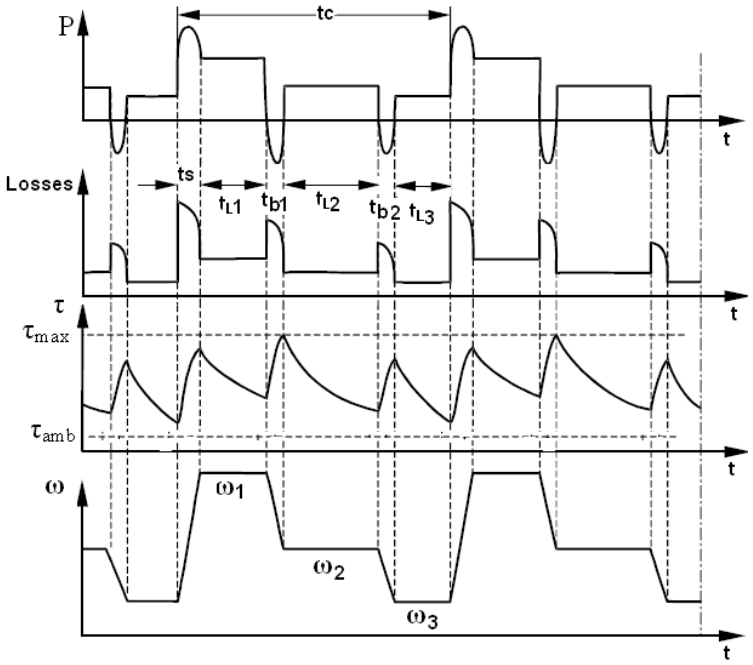


Continuous operation with periodic changes in load and speed (duty type S8): Sequential, identical duty cycles run at constant load and given speed, then run at other constant loads and

speeds. No rest periods. The periods are too short for thermal equilibrium to be obtained during one duty cycle. Speed may change by pole- pair changing method.

$CDF_1=\frac{t_{L1}+t_s}{t_c}.100\%$, $CDF_2=\frac{t_{b1}+t_{L2}}{t_c}.100\%$, $CDF_3=\frac{t_{b2}+t_{L3}}{t_c}.100\%$

C/h=30, 60,120 and 240, FI= 1.2, 1.6, 2, 2.5, 4



DEFINITIONS :

The terms used frequently in intermittent duty drive and hoisting are as below:

Duty : Operation of the motor at load including no load and de-energized period to which the motor is subjected, including the sequence and duration.

Cyclic duration factor : (CDF) : The ratio between the period of loading including starting and electric braking and the duration of the duty-cycle expressed as percentage.

Starting : The process of energizing a motor to bring it up to rated speed from rest.

Jogging or inching (الركض البطيء او الهرولة): This is an incomplete start during which the motor does not attain more than 25% of the rated speed.

Electric braking : A system in which a braking action is applied to an electric motor by causing it to act as a generator.

D.C. Injection braking (dynamic braking) : A form of braking of an induction motor in which a separate dc supply is used to magnetise the motor.

Plug braking (counter-current): A form of electric braking of an induction motor obtained by reversing the phase sequence of its any two lines.

Regenerative braking : A system of electric braking in which energy is returned to the supply system.